

Salaries for San Francisco Employee

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- Role: Data Analyst
- Tools: Python, Pandas, NumPy, Matplotlib and seaborn
- Dataset: salaries for San Francisco Employee

Purpose and scope of the analysis

The purpose of the analysis is to gain a deeper understanding of the salaries for San Francisco and identify key insights that can help drive revenue growth and improve the overall experience.

➤ The scope of analysis including.

- **Data Cleaning:** check null values and duplicate values in every column and after that I renamed the columns.
- **What is the average base pay?**
- **What is the highest amount of Over time Pay in the dataset?**
- **How much does JOSEPH DRISCOLL make (including benefits)? ***
- **What is the name of highest paid person (including benefits)?**
- **What is the name of lowest paid person (including benefits)?**
- **What was the average (mean) BasePay of all employees per year? (2011-2014)**
- **How many unique job titles are there?**
- **What are the top 5 most common jobs?**

What is the average BasePay ?

```
In [9]: sal['BasePay'].mean()
```

```
Out[9]: 66325.44884050643
```

**** What is the highest amount of OvertimePay in the dataset ? ****

```
In [10]: sal['OvertimePay'].max()
```

```
Out[10]: 245131.88
```

**** What is the job title of JOSEPH DRISCOLL ? Note: Use all caps, otherwise you may get an answer that doesn't match up (there is also a lowercase Joseph Driscoll). ****

```
In [15]: sal[sal['EmployeeName']=='JOSEPH DRISCOLL']['JobTitle']
```

```
Out[15]: 24    CAPTAIN, FIRE SUPPRESSION  
         Name: JobTitle, dtype: object
```

**** How much does JOSEPH DRISCOLL make (including benefits)? ****

```
In [16]: sal[sal['EmployeeName']=='JOSEPH DRISCOLL']['TotalPayBenefits']
```

```
Out[16]: 24    270324.91
Name: TotalPayBenefits, dtype: float64
```

**** What is the name of highest paid person (including benefits)?****

```
In [18]: sal[sal['TotalPayBenefits']== sal['TotalPayBenefits'].max()]
```

```
Out[18]:
```

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	TotalPay	TotalPayBenefits	Year
0	1	NATHANIEL FORD	GENERAL MANAGER- METROPOLITAN TRANSIT AUTHORITY	167411.18	0.0	400184.25	NaN	567595.43	567595.43	2011

**** What is the name of lowest paid person (including benefits)? Do you notice something strange about how much he or she is paid?****

```
In [19]: sal[sal['TotalPayBenefits']== sal['TotalPayBenefits'].min()]

#Negative Value
```

```
Out[19]:
```

	Id	EmployeeName	JobTitle	BasePay	OvertimePay	OtherPay	Benefits	TotalPay	TotalPayBenefits	Year
148653	148654	Joe Lopez	Counselor, Log Cabin Ranch	0.0	0.0	-618.13	0.0	-618.13	-618.13	2014

What is the average base pay?

```
In [20]: sal.groupby('Year').mean()['BasePay']
```

```
Out[20]: Year
2011    63595.956517
2012    65436.406857
2013    69630.030216
2014    66564.421924
Name: BasePay, dtype: float64
```

How many unique job title are there?

```
In [4]: sal['JobTitle'].nunique()
```

```
Out[4]: 2159
```

What are the top 5 most common jobs?

```
sal['JobTitle'].value_counts().head(5)
```

```
Transit Operator      7036
Special Nurse         4389
Registered Nurse      3736
Public Svc Aide-Public Works  2518
Police Officer 3      2421
Name: JobTitle, dtype: int64
```

** How many Job Titles were represented by only one person in 2013? (e.g. Job Titles with only one occurrence in 2013?) **

```
In [9]: sum(sal[sal['Year']==2013]['JobTitle'].value_counts() == 1)
```

```
Out[9]: 202
```